Output Beijing time: 2026-05-16 18:15

Figure 1. Study workflow and integrated evidence chain. The workflow summarizes the integrated evidence-chain strategy used to evaluate IL6R in colorectal cancer, including pQTL-based Mendelian randomization, IL6R-locus colocalization, TCGA COAD/READ transcriptomic and reconstructed survival analyses, GSE146771 single-cell analysis, and support-layer network, enrichment and drug-target analyses. The final conclusion emphasizes that MR and colocalization did not support strong genetic causal or shared-variant evidence, whereas downstream transcriptomic and cellular analyses linked IL6R to an immune-, myeloid- and inflammatory-associated tumour microenvironment.

Figure 2. Mendelian randomization analyses of genetically proxied circulating IL6R and colorectal cancer risk. The v39.2/v40 figure summarizes method-specific MR estimates, statistical strength and directionality across the primary FinnGen CRC endpoint and replication CRC GWAS endpoints using readable endpoint labels. The estimates do not provide strong evidence that genetically proxied circulating IL6R has a robust causal effect on colorectal cancer risk.

Figure 3. Colocalization analysis at the IL6R locus. Panel A shows posterior probabilities for coloc hypotheses H0-H4. Panel B provides a compact posterior-evidence summary, and Panel C summarizes the main colocalization conclusion. PP.H4 was approximately 0.048, indicating limited evidence that circulating IL6R protein and colorectal cancer risk share a common causal variant at the IL6R locus.

Figure 4. TCGA expression and reconstructed OS/PFI survival analyses. The v37.3/v40 figure summarizes TCGA patient-level IL6R expression distributions and reconstructed OS/PFI continuous Cox estimates. These analyses support the interpretation that IL6R is not a robust standalone prognostic marker in TCGA colorectal cancer despite its strong tumour-normal expression difference and immune-microenvironment associations.

Figure 5. IL6R-associated immune infiltration patterns in TCGA colorectal cancer. The v37.2/v40 traceable immune-infiltration figure summarizes correlations between IL6R expression and immune/stromal infiltration estimates across deconvolution methods. Dot colour represents Spearman rho and dot size represents statistical strength using a capped -log10(FDR) visual scale; full rho and FDR values are retained in the source tables. The figure emphasizes myeloid, endothelial, neutrophil, macrophage/monocyte and inflammatory tumour-microenvironment features.

Figure 6. Single-cell evidence for IL6-axis activity in GSE146771. The v39.2/v40 single-cell redraw summarizes 10x and Smart-seq2 cell-state structure, IL6-axis module evidence and CellChat-supported communication evidence from traceable source tables. Smart-seq2 provides the clearer IL6-axis communication signal, while the 10x dataset is retained as a sparsity-sensitive comparator. The figure is interpreted as transcriptomic and cell-communication context rather than as independent causal evidence.

Figure 7. IL6R-associated support-layer network, enrichment and drug-target evidence. The v39.3/v40 label-safe figure combines an IL6R-centred support network with immune/inflammatory pathway enrichment and exploratory drug-target support across IL6/IL6R/JAK/STAT and myeloid-associated axes. IL6R-connected edges were retained in the final network rendering. Drug-target annotations are hypothesis-generating and are not interpreted as evidence of clinically validated colorectal cancer therapy.